Hirzebruch–Riemann–Roch theorem

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Contents

In a sense, the introduction will pretend to be accessible, but we shall inevitably have to assume some background in a little bit of everything in geometry and algebra: Riemann geometry, complex and algebraic geometry, Basic K-theory and the theory of characteristic classes, and Elementary Hodge theory. The introduction can be thought of as me trying to make the ideas of the paper as accessible as possible to an undergraduate, but that goal has to drop as soon as we start proving things or else it will the paper will be hundreds of pages of background material.

A natural question to ask in complex and algebraic geometry is "how many holomorphic/meromorphic/algebraic/analytic/etc functions are there on a (suitable) space X". By letting $\mathcal{F}(X)$ represent our choice of functions (the global sections of a [quasi-coherent] sheaf $\mathcal{F}$), we shall denote this question as finding information about the 0th sheaf cohomological group

$$H^0(X, \mathcal{F})$$

which by [1] is equal to $\mathcal{F}(X)$. We denote $\mathcal{F}(X)$ in terms of sheaf cohomology because this group shall sometimes be trivial, and the reason for its trivialness can be described by higher order sheaf cohomology groups:

$$H^1(X, \mathcal{F}), H^2(X, \mathcal{F}), ...$$

exactly how to extract this information is known as the *Riemann Roch Problem.* To motivate the direction the problem took, let us answer this question in some simple cases.

The simplest case would be to deal with polynomial functions over $X = \mathbb{C}$ which are determined by a finite set of roots (counting multiplicity) and a constant:

$$p(z) = c(z - a_1) \cdots (z - a_n)$$

and similarly for rational functions. For future connection to sheaves let $\mathcal{O}_X(X)$ represent all algebraic functions on X and $\mathcal{M}_X(X)$ represent all rational functions on X; other common notation is $\mathbb{C}[X] = \mathcal{O}_X(X)$ and $\mathbb{C}(X) = \mathcal{M}_X(X)$ where $\mathbb{C}$ is often replaced with some other field K or ring R. If we ask for holomorphic functions so that $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{H}(\mathbb{C}))$, then, then the Weierstrass Factorization Theorem (see [9, section 4.2]) tells us that $f \in \mathcal{H}(\mathbb{C})$ factors as an infinite polynomial along with some correction factors

$$z^m e^{g(z)} \prod_k^{\infty} \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2}\left(\frac{z}{a_k}\right)^2 + \cdots + \frac{1}{m_k}\left(\frac{z}{a_{m_k}}\right)^m$$

is the correction factor and the a_k are roots that form a non-accumulating sequence; in particular there are analytic concerns. A meromorphic function f on $\mathbb{C}$ can be determined by two entire function $f = g/h$ (again see [9, section 4.2]); this is possible as for poles of f we can define a function h with the appropriate roots and multiplicities. In other words, determining meromorphic functions becomes a classification of non-accumulating sequences in $\mathbb{C}$. The situation requires even more analytic information in hyperbolic space D^2 (think of the existence of $\sum_n z^{n!}$ and $1/(1-z)$). To understand the algebraic behaviour, limiting the scope to *compact spaces* such as Riemann surfaces or projective varieties gets much better as any holomorphic/meromorphic functions must have finitely many roots/poles by the Identity theorem (that if two functions agree on a set of accumulation points, they are equal), and so we can focus our efforts within the algebraic setting. This setting is plenty rich enough for study and often offers a good starting point to generalize to the non-compact case[1]. In fact, by the efforts of Serre, Chow, and Grothendeick, the category of analytic functions on (suitable) compact spaces is equivalent to the category of algebraic functions on these spaces ([10], [5]), meaning compact spaces are "intrinsically" algebraic. We shall prove some of the pertinent parts of this correspondence in ref:HERE.

Let us try finding holomorphic/meromorphic functions on $X = \mathbb{C}\mathrm{P}^1$ and see if they are determined by the points of $\mathbb{C}\mathrm{P}^1$ just like in the case of $\mathbb{C}$. Immediately, $\mathbb{C}\mathrm{P}^1$ has only constant holomorphic functions by an easy generalization of Liouville or the maximum modulus principle on manifolds. By the Residue Theorem, the sum of the number of roots and poles of a meromorphic function must be 0.

Let us represent this formally. Let

$$D = \sum_i n_i p_i \qquad p_i \in \mathbb{C}\mathrm{P}^1$$

be a finite formal sum; we may think of it as the free (abelian) group $F^{\mathbb{Z}}(\mathbb{C}\mathrm{P}^1)$. Call D a *divisor*, and let $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$ represent this group. A divisor can be thought of as an object that remembers which roots/poles with which multiplicity we are interested in. Define:

$$\mathrm{div}(f) = \sum_{p \in \mathbb{C}\mathrm{P}^1} \nu_p(f) p$$

where $\nu_p(f)$ is the order of the zero/pole, and is negative if it is a pole. This forms a subgroup of $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$; represent it as $\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$ ("P" for principal), and elements of this subgroup are called

[1]Though these cases are still very much being studied, see ref:HERE

principal divisors. This subgroup represents the divisors that are *representable* as a meromorphic function. Let us find which divisors can be represented. Let:

$$\deg D = \sum_i n_i$$

Notice that $\deg(-)$ is certainly a group homomorphism. By our earlier remark:

$$\deg(\mathrm{div}(f)) = 0$$

which is a restriction applied to any meromorphic function on $\mathbb{C}\mathrm{P}^1$. It is easy to see that this restriction would work on any compact Riemann surface (see [9, chapter 6]). If $\deg(D) = 0$, by our earlier remark we have that there exists an f such that $\mathrm{div}(f) = D$, showing this conditions characterizes meromorphic functions $f \in \mathcal{M}(\mathbb{C}\mathrm{P}^1)$ (but it shall not on other Riemann surfaces). Now, consider the *(Weil) divisor class group*

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) = \frac{\mathrm{Div}(\mathbb{C}\mathrm{P}^1)}{\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)}$$

for any $[D] \in \mathrm{Cl}(\mathbb{C}\mathrm{P}^1)$, $\deg([D])$ is well-defined as $\deg(D) = \deg(D + f)$ for any $f \in \mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$. If $\deg(D) = \deg(D')$, then $\deg(D) - \deg(D) = \deg(D - D') = 0$, and hence $D - D' = \mathrm{div}(f)$. Therefore, we have that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}$$

which can be thought of as measuring the obstruction in a meromorphic function being defined by its points/roots. As sheaf cohomology measures obstruction of local information having global information, we shall have:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1})$$

the details of this isomorphism and why $\mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}$ shall be given in ref:HERE. The higher order sheaf cohomology groups shall be trivial as the dimension of $\mathbb{C}\mathrm{P}^1$ is 1 (over $\mathbb{C}$), and hence we'll have:

$$\begin{aligned}
H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) &\cong \mathbb{C}\\
H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \mathrm{Cl}(\mathbb{C}\mathrm{P}^1)\\
H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}(\mathbb{C}\mathrm{P}^1) &\cong \{0\}\\
&\vdots
\end{aligned}$$

Translating the result for $(\mathbb{C}, \mathcal{M}_{\mathbb{C}})$, we see that:

$$\mathrm{Cl}(\mathbb{C}) = \frac{\mathrm{Div}(\mathbb{C})}{\mathrm{PDiv}(\mathbb{C})} \cong 0$$

which shall give us:

$$\begin{aligned}
H^0(\mathbb{C}, \mathcal{O}_{\mathbb{C}}) &\cong \mathbb{C}[x]\\
H^1(\mathbb{C}, \mathcal{O}^*_{\mathbb{C}}) &\cong \{0\}\\
H^2(\mathbb{C}, \mathcal{O}(\mathbb{C}) &\cong \{0\}\\
&\vdots
\end{aligned}$$

as every divisor can be represented by a principal divisor. We can interpret this as there being no impediments to using local information to construct meromorphic functions. Notice that the ring $\mathbb{C}[x]$ is a UFD; this algebraic fact shall be given geometric meaning in ref:HERE.

Let us look at the next simplest nontrivial example. Let $X = \mathbb{C}/\Lambda$ where $\Lambda \subseteq \mathbb{C}$ is a lattice spanned by the $\mathbb{R}$-linearly independent elements $(1, \tau)$ and let $\mathcal{F} = \mathcal{M}_X$ (all rational functions on X); X is called an *elliptic curve*, and so we shall label it as E. As E is compact, if $f \in \mathcal{M}_E$ then $\operatorname{div}(f) = 0$. However, if $\deg(D) = 0$, this does not imply there exists an $f \in \mathcal{M}_E$ such that $\operatorname{div}(f) = D$. To see this, let's analyze $\mathcal{M}_E$ a little more closely. As $\mathbb{C}/\Lambda$ is the quotient by Λ, any meromorphic function must respect the Λ structure:

$$f(z + n + m\tau) = f(z) \qquad \forall n + m\tau \in \Lambda$$

such a function is called *doubly periodic* or *elliptic function* (as they are defined on elliptic curves). Using this periodic condition, it is simple to see that $\operatorname{div}(f) = 0$ by looking at:

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} dz = 0$$

where D is the fundamental domain[2], namely the opposing edges cancel out. We can modify this integral to find other properties, for example:

$$\frac{1}{2\pi i} \int_{\partial D} z \frac{f'(z)}{f(z)} dz = n + m\tau \in \Lambda$$

which is nonzero as z is not doubly periodic. Replacing z by other algebraic functions all end up in Λ, and so we see that if $D = \operatorname{div}(f)$:

$$\sum_i n_i p_i \equiv 0 \bmod \Lambda$$

where here $\sum_i n_i p_i \notin F^{\mathbb{Z}}(E)$ but $\sum_i n_i p_i \in E$, namely we are treating p_i as a complex number multiplied by n_i and summed. Certainly there is no such restriction on a general divisor $D \in \operatorname{Div}(E)$ of degree 0. We thus have shown that there is a surjective map:

$$\operatorname{Cl}^0(E) = \frac{\operatorname{Div}^0(E)}{\operatorname{PDiv}(E)} \twoheadrightarrow \mathbb{C}/\Lambda \tag{1}$$

where $\operatorname{Div}^0(E)$ are all divisor of degree 0. It is in fact an isomorphism, which comes from constructing an elliptic functions using the Weierstrass $\wp$-function, the details are in [9, chapter 5]. Thus, we get the short exact sequence:

$$0 \to \mathbb{C}/\Lambda \to \operatorname{Div}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

often called the *Jacobi-Abel* exact sequence as $\mathbb{C}/\Lambda$ is a Jacobian variety, and a short exact sequence:

$$0 \to \operatorname{Cl}^0(E) \to \operatorname{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

All together, we get a short exact sequence and a commutative diagram:

$$0 \to \operatorname{PDiv}(E) \to \operatorname{Div}^0(E) \to E \to 0$$

[2] If there are zero/poles on the path, we simply shift around appropriately

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Div}^0(E) \arrow[r, hook] \arrow[d, two heads, "\pi"] & \mathrm{Div}(E) \arrow[r, "\deg"] \arrow[d, two heads, "\pi"] & \mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \mathrm{Cl}^0(E) \arrow[r, hook] & \mathrm{Cl}(E) \arrow[r, "\deg"] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Continuing on to other Riemann surfaces, we shall loose the isomorphism given in equation (1) and is the beginning of the study of Jacobian varieties (see [9, section 6.4]). Going in higher dimensions is the beginning of the study of Weil and Cartier Divisors (see [1]), and study how far away a space is from being constructed by "principal" (codimension 1) objects, namely the study of $H^1(X, \mathcal{O}_X)$[3]. Overall, we get some useful information about he obstruction to finding global sections, but not necessarily concrete ways of finding $H^0(X, \mathcal{O}_X)$.

To answer the Riemann Roch problem, a more granular approach to find which functions (of various kinds) exist on a space X must be taken. As divisors keep track of root/pole information, they are a great tool in classifying functions that "satisfy" the properties of the divisor. To make this precise, define an order on divisors $\mathrm{div}(X)$ where (for now) X is either a Riemann surface or a smooth projective curve. Define a partial order on $\mathrm{Div}(X)$ by saying $D_2 \geq D_1$ if each coefficient of $D_2 - D_1$ is non-negative. Then we can consider:

$$\mathcal{O}_X(D)(X) = \{f \in \mathcal{M}_X(X)^* \mid \mathrm{div}(f) + D \geq 0\}$$

In other words, if $D = \sum_i n_i p_i$, then $\mathcal{O}_X(D)(X)$ is all meromorphic functions f where it has a pole at p_i of order at most n_i if $n_i > 0$ and roots at p_i of order at least $|n_j|$ if $n_j < 0$ (and $\mathcal{O}_X(D)$ is a [coherent] sheaf). From this, we can ask the Riemann Roch problem for $H^0(X, \mathcal{O}_X(D))$. The famous Riemann-Roch theorem as proven by Riemann and Roch (where Riemann proved the Riemann-inequality, and Roch provided the necessary work to make it an equality, see [7]) is now known as *Riemann Roch for curves* and is the equality:

$$\dim H^0(X, \mathcal{O}_X(D)) - \dim H^1(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g \tag{2}$$

where g is the genus of the Riemann surface or smooth projective curve (see [11] or [1] for defining the genus of a curve via scheme theory, or for a direct computational proof using normalization see [9]). In the original phrasing of the Riemann Roch theorem, the group $H^1(X, \mathcal{O}_X(D))$ is replaced with $H^0(X, \mathcal{O}(K_X - D))^\vee$ where K_X is the canonical divisor of X; these groups are isomorphic by Serre Duality (see [1, section 6.3]). Then this space is identified with the space of meromorphic 1-forms (either through some simple techniques from complex analysis, see Brill-Noether Reciprocity in either [7] or [9], or by combining of Poincaré duality with DeRham's Theorem, see either [4] or [8]). Switching to the traditional notation $l(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $i(D) = \dim H^0(X, \mathcal{O}_X(K_X - D))^*$ we get:

$$l(D) - l(K_X - D) = \deg(D) + 1 - g \qquad \text{(equiv.)} \qquad l(D) - i(D) = \deg(D) + 1 - g$$

This result is proven in theorem 0.4.1. As we phrased the Riemann-Roch problem in the beginning, this formula would then be asking for the value of $l(D)$. There are cases where $l(K - D)$ disappears (for example if $D = 0$ or $D = K$), and there are many cases we can compute it easily enough. It is simple to compute the right hand side but in general it is harder to computer the left hand side. Thus, this formula becomes a new constriction that helps us find $l(D)$. There are a set of theorems known as *vanishing theorems* that specify under which conditions $l(K - D)$ (or more generally, higher cohomology groups) vanish (see [6]); under these conditions we exact recover $l(D)$.

[3] Notice the object are codimension 1, so the "dual" have dimension 1

Let us work towards the intuition behind the generalization. The notation on left hand side in equation (2) can be compactified by using the Euler-Poincaré characteristic (or simply the Euler characteristic): $\chi(X, \mathcal{O}_X(D)) = \sum_{i=0}^{\infty}(-1)^i \dim H^i(X, \mathcal{O}_X(D))$ so that:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g$$

As X is a curve $H^i(X, \mathcal{O}_X(D)) = 0$ for $i > 1$, cite? Reference for later? giving the equality. The right hand side shall be drastically different, split into special invariants of vector bundles known as the Chern class and Todd class mention the "duality" between them? see notes . In fact, the Euler Characteristic shall also turn out to be an invariant on vector bundles, and hence the Riemann Roch theorem generalizes to a question about the existence of sections on vector bundles over a compact manifolds, namely if $(X, \mathcal{F})$ is a compact manifold along with a vector bundle $\mathcal{F}$, then we want to compute:

$$H^0(X, \mathcal{F})$$

which leads to finding

$$\chi(X, \mathcal{F}) = \int_X \mathrm{ch}(X)\mathrm{td}(X)$$

This equation is precisely the result of the *Hirzebruch-Riemann Roch Theorem*. We shall mention the generalization to smooth quasi-projective schemes over a field near the end, and also talk a bit about a recent paper ([2]) generalizing to finite dimensional Noetherian schemes.

To motivate how vector bundles will enter the picture, it is worth-while seeing how $\mathcal{O}_X(D)$ can be interpreted as a line bundle. To that end, recall that the projective curve $X = \mathbb{P}^1$ as defined in classical algebraic geometry only has constant global sections. Namely, if $U_0, U_1 \cong \mathbb{A}^1$ is the usual cover with global sections (i.e. functions) $\mathcal{O}_X(U_0) \cong \mathcal{O}_X(U_1) = K[x]$ and the compatibility condition on $U_0 \cap U_1$ is $f(z) = g(1/z)$, namely:

$$\begin{tikzcd} U_0 & U_1 \\ U_0 \cap U_1 \arrow[u, hook] \arrow[r, "\cong"] & U_0 \cap U_1 \arrow[u, hook] \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} \mathbb{A}^1 & \mathbb{A}^1 \\ \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \arrow[r, "\cong"] & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \end{tikzcd}$$

gives:

$$\begin{tikzcd} \mathcal{O}_X(U_0) \arrow[d] & \mathcal{O}_X(U_1) \arrow[d] \\ \mathcal{O}_X(U_0 \cap U_1) \arrow[r, "\cong"] & \mathcal{O}_X(U_0 \cap U_1) \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} K[x] \arrow[d] & K[y] \arrow[d] \\ K[x, 1/x] \arrow[r, "\cong"] & K[y, 1/y] \end{tikzcd}$$

where $x \mapsto 1/y$ and $1/x \mapsto y$, Then if $F([x' : y']) \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is a global function where $x = x'/y'$ and $y = y'/x'$ (when $y', x' \neq 0$ respectively), then $F([x : 1]) \to f(x) \in \mathcal{O}_X(U_0)$ and $F([1 : y]) \to g(y) \in \mathcal{O}_X(U_1)$ are the restrictions of the function to U_0 and U_1 respectively, and by the above diagram it must be that these functions map to the same element in the restriction, namely:

$$f(z) \to f(x) = f(1/y) = g(y) \leftarrow g(y)$$

but it is impossible that $f(1/y) = g(y)$ unless f, g are the same constant. Thus, it must be that the only global functions on the projective line are constants:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = K$$

The fundamental limitation here is that the restriction $f(x) = g(1/x)$ is very strong. What if we can loosen this restriction? For example what if

$$f(x) = xg\left(\frac{1}{x}\right)$$

Then we can in fact form linear homogeneous functions! For example consider

$$F([x' : y']) = ax' + by'$$

Then F maps to $f(x) = ax + b$ and $g(y) = a + by$. Then:

$$f(x) = ax + b = x\left(a + b\frac{1}{x}\right) = xg\left(\frac{1}{x}\right)$$

which works! More generally, if the condition $f(x) = x^n g(1/x)$ allows for degree n homogeneous polynomials, and the homogeneity can happen for any number of variables. Denote the degree n-homogeneous polynomials for $\mathbb{P}^k$ by $\mathcal{O}_{\mathbb{P}^k}(n)(\mathbb{P}^k)$.

Now, recall that a n-dimensional vector bundle $\pi : E \to X$ over a field K can be defined by taking open sets $U_i \subseteq X$ such that $\pi^{-1}(U_i) \to U_i \times F$ and insuring that for any intersection $U_i \cap U_j$ we have a gluing map:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to \mathrm{GL}_n(K)$$

such that $g_{jk}g_{ij} = g_{ik}$ (the cocycle condition) which specifies the gluing condition. When limiting to line bundles, the gluing map becomes:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to K^*$$

Intuitively, this give a point-wise gluing conditions on the bundle: if $v \in U_i$ and $w \in U_j$ that are the same point in $U_i \cap U_j$, then:

$$v = A_{ij}w$$

Moving onto the construction we have done earlier, as we are saying that $\mathcal{O}_{\mathbb{P}^1}(n)$ ought to be considered as a line bundle, we see that we ought to consider:

$$g_{ij} : \mathcal{O}_{\mathbb{P}^1}(U_i \cap U_j) \to \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j) \cong k[x, 1/y] \setminus \{0\}$$

where we take $\mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$ in place of K^* as these are the "units" (i.e. isomorphisms) on this intersection. For example, given f, g restrict to the same functions on the intersection, then we now "twist" g to be:

$$f(x) = x^n g(1/x) \qquad x^n \in \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$$

The word "twist" comes from both complex analysis where z^n twists $\mathbb{C}$ onto itself around the branching point, and from algebraic geometry where the non-triviality of bundles can be interpreted by characteristic classes as some non-trivial twisting (we shall cover this intuition in ref:HERE). The details of this correspondence is given in [1, section 4.1] (especially how to transition from points of a bundle to section of a sheaf). Overall, we see that $\mathcal{O}_{\mathbb{P}^1}(n)(\mathbb{P}^1)$ is the global sections of a line bundle; more generally $\mathcal{O}_{\mathbb{P}^1}(n)$ remember all the sections over each open set U, as $\mathcal{O}_{\mathbb{P}^1}(n)$ is a sheaf.

With this intuition, we can see how to interpret $\mathcal{O}_X(D)$ as a line bundle. Choosing an affine cover U_i for X, we can define:

$$D|_{U_i} = \sum_{p \in U_i} n_p p$$

which is the representation of the divisor D in U_i. Then for each affine U_i, choose $s_i \in \mathcal{M}_X(U_i)$ so that $\langle s_i \rangle_K = \mathcal{O}_X(D)(U_i)$ is a generator as a K-algebra of the local sections and:

$$\operatorname{div}(s_i) = -D|_{U_i}$$

So that s_i has roots/poles of the appropriates order. Such a section exists, for $U_i \cong \mathbb{A}^1$ (as X is a curve) so $\mathcal{O}_X(U_i) \cong k[x]$ and so by our earlier discussion we can certainly find an s_i with exactly zeros and poles at $-D|_{U_i}$ and it is an exercise in ring-theory to show this is a generator (ex. think of choosing a uniformization parameter for a local ring).

Let us label the collection of the meromorphic functions $\mathcal{O}_X(D)(U_i)$. To find an $f \in \mathcal{O}_X(D)(X)$ (i.e. a global section), we need to insure we can glue together nicely on intersection. Consider (nonempty) $U_i \cap U_j$ and take the generators $s_i \in \mathcal{O}_X(D)(U_i)$ and $s_j \in \mathcal{O}_X(D)(U_j)$. It is not hard to see that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. Then certainly on $U_i \cap U_j$:

$$\operatorname{div}(s_i) = \operatorname{div}(s_j) = -D|_{U_i \cap U_j}$$

and so, as we are considering meromorphic functions, we can consider:

$$\operatorname{div}\left(\frac{s_j}{s_i}\right) = \operatorname{div}(s_j) - \operatorname{div}(s_i) = 0$$

As s_j/s_i has no zero's/pole on $U_i \cap U_j$, this means that it is holomorphic and nowhere vanishing on $U_i \cap U_j$. But then:

$$g_{ij} = \frac{s_j}{s_i} \in \mathcal{O}_X^*(U_i \cap U_j)$$

But then we have the desired transition functions! Thus, we may glue them to form all global sections $\mathcal{O}_X(D)(X)$for every D, and then $\mathcal{O}_X(D)$ is a line bundle. In most cases (and in most spaces that will be considered in this paper), every line bundle $\mathcal{L}$ is induced by a divisor D; [1, section 4.3] shows how there is a bijective correspondence between line bundles and (Weil/Cartier) divisors on the appropriate type of scheme.

This also means that $\deg(D)$ should be interpreted as a property of a bundle. (use this to build up to Chern classes).

0.1 A bit of Hodge Theory

Let X be a complex analytic manifold of complex dimension n. As a real manifold ,it is a smooth manifold of dimension $2n$. Let TX be the tangent bundle and $T_{X,x}$ the fiber at x, i.e. the $\mathbb{C}$-vector space of (complex) dimension n. Let $z_1, ..., z_n$ be complex local coordinates at $x \in X$; these are associated to real coordinates $x_1, y_1, ..., x_n, y_n$ where $z_j = x_j + iy_j$. Then an $\mathbb{R}$-basis at $T_{X,x}$ is

0.2 GAGA results

Define Z-topology and $\mathbb{C}$-topology on a variety/closed manifold X.

0.3 Characteristic Classes

here

0.4 Hirzebruch-Riemann-Roch

(This may not be important for us; it will give the fact that $\dim_{\mathbb{C}} H^i = 0$ for certain i.)

X is admissible if:

1. X is σ-compact (locally compact and a countable union of compacts)
2. X has finite combinatorial dimension X

As (1) implies paracompact, our work on fine and soft sheaves goes through.

Theorem 0.4.1: RR For Riemann Surface with Line Bundles

Let X be a compact Riemann surface and $\mathcal{L}$ a complex analytic line bundle on X. Then

1. There exists a Cartier divisor D such that $\mathcal{L} \cong \mathcal{O}_X(D)$ so that $\deg(\mathcal{L}) := \deg(D)$
2. The Riemann-Roch equation holds:

$$\dim_{\mathbb{C}} H^0(X, \mathcal{L}) - \dim_{\mathbb{C}} H^0(X, \omega_X \otimes \mathcal{L}^\vee) = \deg(\mathcal{L}) + 1 - g$$

where (by ref:HERE, or [1, section 5.3]) $g = \dim H^0(X, \omega_X) = \dim H^1(X, \mathcal{O}_X)$ is the genus of X.

Proof **:**
The first point is an application of much proved before: By ref:HERE X is an algebraic variety, in particular a curve, and by [1, section 6.1] there is an embedding $X \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$ for some N. Then by ref:HERE (GAGA, which I'll prove) we have that $\mathcal{L}$ is isomorphically pushed forward through these maps hand is an algebraic line bundle. As X is a variety, by [1, section 4.3] there exists a Cartier divisor D such that $\mathcal{L} = \mathcal{O}_X(D)$. Let us show $\deg(\mathcal{L}) = \deg(D)$ is well-defined, i.e. independent of linear equivalence. Let $f \in \mathcal{M}(X)$. Then as shown in [9, section 6.3] there exists a branch covering $f : X \to \mathbb{P}^1_{\mathbb{C}}$ where:

$$\#(f^{-1}(0) = \#(f^{-1}(\infty)) = \text{deg. of map}$$

Hence:

$$\deg(f)\#(f^{-1}(0) - \#(f^{-1}(\infty)) = 0$$

Thus, $D \sim E$ implies $\deg(D) = \deg(E)$, hence $\deg(\mathcal{L})$ is well-defined

For the 2nd point, if $D = 0$, then $\mathcal{O}_X(0) = \mathcal{O}_X$, and $H^0(X, \mathcal{O}_X) = \mathbb{C}$ so $l(0) = 1$ and $i(0) = g$ and

$$l(0) - i(0) = 1 - g = \deg(0) + 1 - g$$

by [9, section 6.3], hence assume $D \neq 0$ (though the beginning of this discussion doesn't need this assumption). First use Serre Duality to get:

$$H^0(X, \omega_X \otimes \mathcal{L}^\vee) \cong H^1(X, \mathcal{L})^\vee$$

Reducing the right hand side to the Euler characteristic $\chi(X, \mathcal{O}_X(D))$. To get the right hand side, we take advantage of additivity of χ. For any point $P \in X$, we get an exact sequence:

$$0 \to \mathcal{O}_X(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

when κ_P is the skyscraper sheaf:

$$(\kappa_P)_x = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

Note that as this sheaf is flasque, H^i disappears for $i > 0$ so $\chi(X, \kappa_P) = 1$.

Go over $\chi(X, \kappa_D)$

Tensoring with $\mathcal{O}_X(D)$, and applying our theory from [1, section 4.3] we get:

$$0 \to \mathcal{O}_X(D - P) \to \mathcal{O}_X(D) \to \kappa_P \otimes \mathcal{O}_X(D) \to 0$$

Thus, by additive of χ:

$$\chi(X, \mathcal{O}_X(D) = \chi(X, \kappa_P \otimes \mathcal{O}_X(D)) + \chi(X, \mathcal{O}_X(D - P))$$

Now, if $D > 0$, we can pick a point P appearing in D. Then $\deg(D-P) = \deg(D) - 1$. We can thus do induction on $\deg(D)$. If $\deg(D) = 0$, then $\mathcal{O}_X(D)$ has holomorphic functions, which as X is compact is only constant functions, and by the Brill-Noether Reciprocity ([9]) we have $i(D) = g$, giving the base case. Now suppose the result holds for $\deg(D) = n - 1$ and consider $\deg(D) = n$. Then $\deg(D - P) = n - 1$ from which we get by the inductive hypothesis

$$\deg(D - P) + 1 - g = \chi(X, \mathcal{O}_X(D - P))$$

words here; I need χ to split with D and P

$$\deg(D) + (-1 + 1) + 1 - g = \chi(X, \mathcal{O}_X(D))$$

giving the desired result.

For arbitrary D, we can split $D = D^+ - D^-$ where $D^+, D^- \geq 0$. Then we get a short exact sequence:

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(D^-) \to \kappa_{D^-} \to 0$$

Then by additivity;

$$\chi(X, \mathcal{O}_X(D^-)) = \chi(X, \mathcal{O}_X) + \chi(X, \kappa_{D^-}) = 1 - g + \deg(D^-)$$

Now, tensor with $\mathcal{O}_X(D)$ to get the short exact sequence:

$$0 \to \mathcal{O}_X(D) \to \mathcal{O}_X(D + D^-) \to \kappa_{D^-} \to 0$$

where the last term is the same (as can be verified point-wise). Applying additivity of χ again we get:

$$\chi(X, \mathcal{O}_X(D + D^-)) = \chi(X, \mathcal{O}_X(D)) + \chi(X, \kappa_{D^-}) = \chi(X, \mathcal{O}_X(D)) + \deg(D^-)$$

As $D + D^- = D^+$, re-arranging we get:

$$\chi(X, \mathcal{O}_X(D)) = -\deg(D^-) + \chi(X, \mathcal{O}_X(D^+))$$

Applying our earlier result we get:

$$\chi(X, \mathcal{O}_X(D)) = -\deg(D^-) + \deg(D^+) + 1 - g \deg(D) + 1 - g$$

as we sought to show.

As $\mathcal{L}$ is a line bundle, we can take its value of the first Chern class $c_1(\mathcal{L}) \in H^2(X;\mathbb{Z})$. By letting $[X] \in H - 2(X;\mathbb{Z}) \cong \mathbb{Z}$ be the fundamental class, then:

$$\deg(\mathcal{L}) = c(\mathcal{L})[X] \in \mathbb{Z}$$

Then theorem 0.4.1 becomes:

$$\begin{aligned}\chi(X, \mathcal{L}) &= c_1(\mathcal{L})[X] + 1 - g \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}(2 - 2g)\right][X] \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}c_1(T_X^{1,0})\right][X]\end{aligned}$$

By Bass-Serre Cancellation (see [3, section 5.1]) or by some result from the theory of characteristic (see [3, section 3.2]), higher-dimensional vector bundles are trivial; in particular as $\dim_{\mathbb{C}} X = 1$, any vector bundle of dimension $n > 1$ will have "trivial" information cohomologically. Let us show this explicitly to further motivate HRR. We first require a special case of

This following theorem may also be in my char.class notes

we also technically don't need it as we can prove the result for vector bundles via HRR with a lot more theory, but I think this is good to confirm that HRR is correct

Theorem 0.4.2: Atiyah-Serre On Vector Bundles

Let X be either a compact complex manifold or an algebraic variety; in either case of dimension $d = \dim_{\mathbb{C}} X$. Let E be a rank $r > d$ vector bundle on X that is smooth or algebraic whether X is a manifold of variety respectively [a]. Then there is a trivial bundle $\mathbb{I}^{r-d}$ of rank $r - d$ that fits into the short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

and $\mathrm{rk}(E'') = d$.

[a] If algebraic, assume $\Gamma_{\mathrm{alg}}(X, \mathcal{O}_X(E)) \to E_x$ is surjective for all x;i.e. it is generated by its global sections

Proof **:**
If E is smooth, then by partitions of unity it is always generated by its global sections. As algebraic vector bundles need not be fine, we must assume they are (for example $\mathcal{O}_{\mathbb{P}^1}(-1)$ does not satisfy this condition). Without loss of generality, we shall work over $\overline{k} = \mathbb{C}$.

Then, for each $x \in E$, $\Gamma(X, \mathcal{O}_X(E)) \twoheadrightarrow E_x$ surjective onto the fiber E_x. Then we can choose a finite dimensional subspace $W(x) \subseteq \Gamma(X, \mathcal{O}_X(E))$ as $W(x) \to E_x$ surjective. Then either $\mathbb{C}$- or Z-near x, $\Gamma(X, \mathcal{O}_X(E))$ surjects onto the fibers in the neighbourhood (namely, the determinant is smooth/regular, and hence we can follow a similar argument in both cases). As X is [quasi]-compact, there exists a finite open cover with this property.

Label these choices $W(i)$. Taking all their generators, we have a finite dimensional subspace $W \subseteq \Gamma(X, \mathcal{O}_X(E))$ such that $W \twoheadrightarrow E_x$ surjects for all $x \in X$ (namely $\sigma \mapsto \sigma(x)$ surjects). Let:

$$\ker(x) = \ker(W \to E_x)$$

Then as $\dim E_x = r$,

$$\dim \ker(x) = \dim W - r$$

As the dimension is constant, $\dim \ker(x)$ is independent of X, thus we can take their union of X. To be precise, let $\mathbb{P}(\ker(x)) \subseteq \mathbb{P}(W)$ be the projectivizations of $\ker(x) \subseteq W$ (i.e. collection of all lines). Then consider

$$Z = \cup_{x \in X} \mathbb{P}(\ker(x))$$

and takes its Z-closure if necessary. Then:

$$\dim Z = \dim X + \dim W - r - 1 = \dim W + d - r - 1$$

where the -1 comes from the fact that $\dim(\mathbb{P}(V)) = \dim V - 1$. Hence, we have:

$$\operatorname{codim}(Z \hookrightarrow \mathbb{P}(W)) = r - d$$

Then it is simple algebraic/affine geometry to see there exists a projective (linear) subspace $T \subseteq \mathbb{P}$ with $\dim T = r - d - 1$ such that

$$T \cap Z = \emptyset$$

Let $T = \mathbb{P}(S)$ for some subspace $S \subseteq W$ (so $\dim S - r - d$). Then by construction:

$$X \times S = X \times \mathbb{C}^{r-d} = \mathbb{I}^{r-d}$$

From this, we can define a map $\mathbb{I}^{r-d} \hookrightarrow E$ sending $(x, s) \in s(x) \in E$. As $T \cap Z = \emptyset$, $s(x)$ is never zero, and hence $\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$ has full rank at each point. Letting $E'' = E/\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$, we get a vector bundle of dimension d and a short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

as we sought to show.

Now, by section 0.3 we have that $c_i(\mathbb{I}^r) = 0$ for all i, so by additivity of characteristic classes we have that $c_1(E) = c_1(E'')$. As $c_1(E) = c_1\left(\bigwedge^r E\right)$, we get:

$$c_1(E) = c_1\left(\bigwedge^{\mathrm{rk}(E'')} E''\right)$$

Using this, we can prove the following "simple" generalization:

Theorem 0.4.3: RR For Riemann Surface With Vector Bundles

Let X be a compact Riemann surface and E a complex vector bundle over X of rank r. Then:

$$\chi(X, E) = c_1(E) + r(1 - g)$$

whee $c_1(E)$ is the first Chern class of E.

Proof **:**

Start with embedding X in $\mathbb{P}^N$ via an algebraic equation and pushforward E so that it is an algebraic vector bundle. Then E need not be generated by global holomorphic functions, however by twistign it enough

$$E(n) := E \otimes \mathcal{O}_X(n) \cong E \otimes \mathcal{O}_X^{\otimes n}$$

we shall have a very ample sheaf, and hence $E(n)$ shall have enough global section (see [1, section 4.4] for the details). From this, we can apply theorem 0.4.2:

$$0 \to \mathbb{I}^{r-1} \to E(n) \to E'' \to 0$$

where $\mathrm{rk}(E'') = 1$. Then twisting it by $\mathcal{O}_X(-n)$ we get:

$$0 \to \coprod_{r-1} \mathcal{O}_X(-n) \to E \to E''(-n) \to 0 \tag{3}$$

Then:

$$\chi(X, \mathcal{O}_X(E)) = \chi(X, E''(-n)) + \chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right)$$

As $\mathrm{rk}(E''(-n)) = 1$, by ordinary Riemann-Roch we have:

$$\chi(X, E''(-n)) = c_1(E''(-n)) + 1 - g$$

giving us:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E''(-n)) + 1 - g + \chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right)$$

From this, we shall proceed with inducting on the rank r. The base case $r = 1$ was theorem 0.4.1. Assume Riemann Roch holds for $r - 1$, so that by the inductive hypothesis:

$$\chi\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) = c_1\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) + (r-1)(1-g)$$

Then:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E''(-n)) + 1 - g + c_1\left(\coprod_{r-1} \mathcal{O}_X(-n)\right) + (r-1)(1-g)$$

Then we can combine the Chern classes by additivity on equation (3), and so;

$$\chi(X, \mathcal{O}_X(E)) = c_1(E) + r(1-g)$$

as we sought to show.

We shall show that:

$$\chi(X, \mathcal{O}_X(E)) = c_1(E) + \frac{\mathrm{rk}(E)}{2} c_1(T_X^{1,0})$$

0.5 Multiplicative sequecnes

As we saw with Chern and Pontryagin classes, their product translates to a cup product. This gives a natural algebraic relation that we know codify for future reference. Let B be a commutative ring, and define:

$$\mathcal{B} = B[z, p_1, p_2, ...]$$

Define a grading given by the following:

$$p_{j_1} p_{j_2} \cdots p_{j_r} \text{ has weight } j_1 + j_2 + \cdots + j_r$$

Then let $\mathcal{B}_k$ be the weight k-groups; note $\mathcal{B}_0 = B$ and $\mathcal{B}_r \mathcal{B}_s = \mathcal{B}_{r+s}$. Let $\pi(k)$ be all partitions of k. Then $\mathcal{B}_k$ is a B-module of rank $\pi(k)$.

Definition 0.5.1: Multiplicative Sequence

Let $\{K_j\}$ be a sequence of polynomials in indeterminates p_i where $K_0 = 1$ and $K_j \in \mathcal{B}_j$. Then the sequence $\{K_j\}$ is called a *multiplicative sequence* if whenver a power series is the product of two power series:

$$1 + p_1 z + p_2 z^2 + \cdots = (1 + p_1' z + p_2' z^2 + \cdots)(1 + p_1'' z + p_2'' z^2 + \cdots)$$

with indeterminates z, p_i', p_i'', then we get the identify:

$$\sum_{j=0}^{\infty} K_j(p_1, p_2, ...p_j) z^j = \left(\sum_{i=1}^{\infty} K_i(p_1', p_2', ...p_i') z^i\right)\left(\sum_{j=1}^{\infty} K_j(p_1'', p_2'', ...p_i'') z^j\right)$$

that is:

$$K\left(\sum_{i=0}^{\infty} q_i z^i\right) = \sum_{i=0}^{\infty} K_i(q_1, ..., q_i) z^i$$

where $q_i \in \mathcal{B}$ of weight i. In other words a sequence $\{K_j\}$ is called multiplicative if it induces an endomorphism on

$$B[p_1, p_2, ...]\,[[z]]$$

Given a multiplicative sequence $\{K_j\}$, we associate to it its *characteristic power series* as the image of $K(1+z)$:

$$K(1+z) = \sum_{j}^{\infty} K_j(1,0,...,0)z^j = \sum_{j}^{\infty} b_j z^j$$

Recall we can define Chern roots. Hence we will be considering:

$$1 + p_1 z + \cdots + p_n z^n = \prod_{i}^{n} (1 + \beta_i z)$$

where β_i is a symmetric polynomial in the p_i's. By some elementary ring theory and theory of symmetric functions, we can interpret $\mathcal{B}$ as the ring of all symmetric functions in $\beta_1, ..., \beta_n$.

Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[2] A. Navarro and J. Navarro. "On the Riemann-Roch formula without projective hypothesis". May 30, 2017. DOI: 10.48550/arXiv.1705.10769. URL: http://arxiv.org/abs/1705.10769.

[3] Nathanael Chwojko-Srawley. *Everything You Need to know about K-theory*. Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_K-Theory.pdf.

[4] John M. Lee. *Introduction to Riemannian Manifolds*. Vol. 176. Graduate Texts in Mathematics. Springer International Publishing, 2018. ISBN: 978-3-319-91754-2 978-3-319-91755-9. URL: http://link.springer.com/10.1007/978-3-319-91755-9.

[5] Wei-Liang Chow. "On Compact Complex Analytic Varieties". In: 71 (4 1949), pp. 893–914. ISSN: 0002-9327. DOI: 10.2307/2372375. URL: https://www.jstor.org/stable/2372375.

[6] Pierre Deligne and Luc Illusie. "Relèvements modulop2et décomposition du complexe de de Rham". In: 89 (2 June 1, 1987), pp. 247–270. ISSN: 1432-1297. DOI: 10.1007/BF01389078. URL: https://doi.org/10.1007/BF01389078.

[7] Phillip A. Griffiths. *Introduction to Algebraic Curves*. Vol. 78. Mathematical Monographs. American Mathematical Society.

[8] Nathanael Chwojko-Srawley. *Everything You Need To Know About Differential Geometry*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_differential_geometry.pdf.

[9] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.

[10] Jack Hall. "GAGA theorems". May 17, 2022. DOI: 10.48550/arXiv.1804.01976. URL: http://arxiv.org/abs/1804.01976.

[11] Robin Hartshorne. *Algebraic geometry*. Vol. 52. Springer Science & Business Media, 2013. URL: https://books.google.com/books?hl=en&lr=&id=7z4mBQAAQBAJ&oi=fnd&pg=PR13&dq=info:XErvue4ovgwJ:scholar.google.com&ots=OR51pvoipT&sig=NcjWQsiJwE9XL79fBWQTMMub5WM.